

Quantitative morphology and network representation of soil pore structure

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Abstract

Pore-network models are attractive to relate pore geometry and transport processes in soil. In this contribution a ‘morphological path’ is presented to generate a network model based on quantitative morphological investigations of the 3D pore geometry in order to predict soil hydraulic properties. The 3D-geometry of pores larger than 0.04 mm in diameter is obtained using serial sections through impregnated samples. Beside pore-size distribution an important topological aspect of pore geometry is the spatial connectivity of the pore space which is difficult to measure. A connectivity function is proposed defined by the 3D-Euler number. The goodness in the estimation of the Euler number using serial sections is investigated in subsamples of different sizes and shapes. Then, a simple network model is generated which can be adapted to a predefined pore-size distribution and connectivity function. Network simulations of hydraulic properties are compared to independent measurements at the same soil material and the effect of topology on water flow and solute transport is investigated. It is concluded that a rough estimation of pore-size distribution and topology defined by the connectivity function might be sufficient to predict hydraulic properties. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Water flow and solute transport through soil are directly related to the geometry of the available pore space. In the last two decades, network models have been explored in the field of soil physics to study processes at the pore scale. For a recent review see [1].

Network models are idealized representations of the complex pore geometry allowing for an efficient calculation of water flow. By varying different geometric aspects of the network and evaluating the corresponding effect on water flow and solute movement our knowledge of the relation between pore structure and transport properties has increased considerably. Jerauld and Salter [2] investigated the effect of the spatial correlation of pore size on permeability as well as on the phenomenon of hysteresis of the capillary pressure–saturation relation. The spatial heterogeneity of pore-size distribution was studied by Ferrand and Celia [3] and the effects of anisotropy by Friedman

and Seaton [4]. The effect of constrictions within the pore space, meaning the topological characteristics, appeared to play a central role in the hysteretic behavior of hydraulic properties. Such geometries are typically modeled by relatively large pore bodies at the nodes of the network which are connected by smaller throats.

It is still a challenge to use network models as a predictive tool for water flow and solute transport. The critical point is how to get the geometric parameters to construct the network. One approach which has been successfully applied to predict the unsaturated hydraulic conductivity [5,6] and air permeability [7] is to fit the parameters of the network model to an experimentally measured pressure–saturation relation. However, there is no unique solution since the pressure–saturation relation depends on both the pore-size distribution and the pore topology [8]. Another approach is to measure the geometrical characteristics of the pore morphology directly using undisturbed samples. This approach is attractive because

(i) pore morphology is, in principle, directly observable without any experiment;

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(ii) the morphology of the pore space can be interpreted in terms of processes of structure formation and hence, these processes may be directly related to flow and transport in soil.

To follow the path of the morphological approach different steps are required. First the undisturbed structure of the material must be obtained. This can be done using serial sections through impregnated samples [9] or by non-invasive techniques as X-ray tomography [10]. Then, the complex geometry of the pore space must be quantified in terms of suitable parameters, which in a following step can be realized in a network model. A crucial question is, what are the suitable parameters?

From the previous studies cited above it can be concluded that, at the pore scale, there are mainly two controlling factors for soil hydraulic properties and solute transport: The pore-size distribution and the topology of the pore space meaning the way in which pores of different sizes are interconnected. Thereby, the microscopic shape of pore–solid interfaces and the physico-chemical properties of the solids are ignored. Moreover, the pore structure is considered to be static. This is assumed to be reasonable for mineral soils with low carbon and low clay content.

Most of the techniques used on the morphological path – sample preparation, digital imaging, measurement of pore-size distribution – are straightforward and are discussed here only briefly. The focus of this paper lies on the quantification of topological characteristics, and how to incorporate such information into a network model. Although topology is highly significant in terms of flow and transport in porous media, it is rarely measured directly. In typical network models, topology is implicitly determined within narrow bounds by the choice of the network configuration, i.e. the arrangement of bodies and throats and certain correlation rules.

Using 3D representations of the complex pore geometry, a direct measure of pore topology is introduced, termed the connectivity function [11]. It is defined by the 3D-Euler number of the pore space in dependency of the pore radius. Then, a simple, hierarchical network model is proposed which can be adapted to a predefined connectivity function. Thereby, the concept of pore bodies and pore throats is dropped, however, the effect of constrictions as realized by a body–throat formulation of the model is preserved. As a consequence we gain more flexibility in terms of pore continuity. In a classical body–throat model, large pores at the nodes are always connected through narrow throats. This may be an excellent model for granular media as sandstones. In soil, however, there are large pores, which are continuous over a considerable length, i.e. root channels or burrows of animals. These pores can be more easily represented by the model proposed here. As another aspect of pore geometry, the size spectrum of pores is considered. In soil, the size spectrum of pores is far too large (sub-

micrometer–millimeters) to be represented by a single network model. Consequently, pores within a network model are expected to be connected through smaller pores which are not explicitly represented by the model. This may be an important difference for soil compared to other consolidated materials as rocks or sandstones, where network models have been introduced [12]. In the model proposed here, ‘small pores’ referred to as ‘matrix porosity’ are represented by an effective pore radius.

The morphological path to generate a simple but highly flexible network model was demonstrated for a silty top soil by Vogel and Roth [13]. In this paper, the predictive power of the network model based on morphological measurements is tested for the same soil by analyzing a number of realizations of the network model. This allows to investigate the variability of the results introduced by the randomness of those parameters which cannot be measured directly. Moreover, the effect of network size on the variability of different properties as capillary pressure–saturation relation, hydraulic conductivity and solute transport is studied. The simulated results of hydraulic properties are compared to experimental findings. Additionally, the quality of the topological characterization of the porous medium is investigated by analyzing the variance obtained for subsamples according to Vogel [14] who demonstrated this approach for another soil.

2. Quantitative morphology

2.1. Sample preparation

The soil used for this study was a silty agricultural top soil (Orthic Luvisol) near Jülich (Germany). It is the same soil as investigated by Vogel and Roth [13], where a detailed description of the sampling procedure is given. To investigate the pore structure, six undisturbed samples were impregnated with a polyester resin. Subsequently, each sample was cut into a stack of 20 vertical serial sections (20 mm × 13 mm) with a separation of 0.04 mm. The sections were photographed using a digital camera at a resolution of 1523 × 1011 pixel and 0.013 mm/pixel. Another series of 20 serial sections was produced at a lower resolution (35 mm × 23 mm, 0.023 mm/pixel) at each location to capture larger pores more representatively. In this way pores larger than 0.04 mm in diameter could be measured. The 3D-geometry of the pore space was reconstructed from the digitized images of the serial sections. Hence, the resulting volume was 20 mm × 13 mm × 0.76 mm and 35 mm × 23 mm × 0.44 mm, respectively. Fig. 1 shows the binary image of pores in one serial section. The 3D-reconstruction of a small part of the corresponding series of 20 serial sections is shown in Fig. 2.

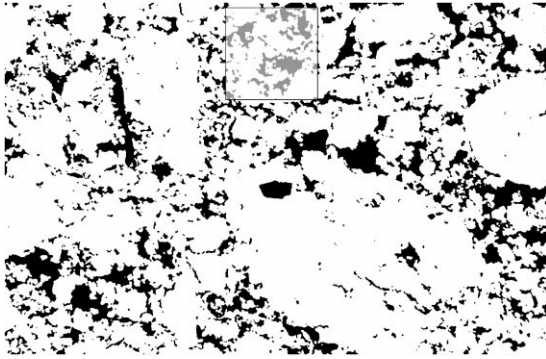


Fig. 1. Serial section of the pore structure (20 mm × 13 mm), pores are black. The grey-shaded part is used for 3D-reconstruction in Figs. 2 and 3.

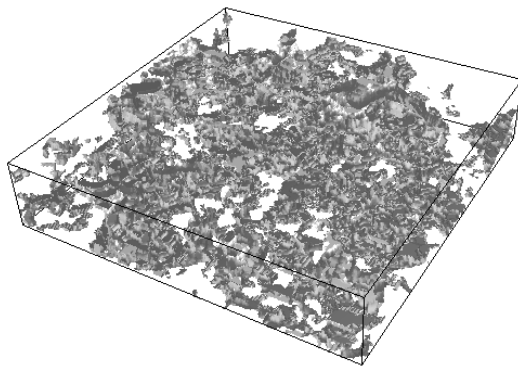


Fig. 2. 3D-reconstruction of pores >0.04 mm within a subsample of 3.25 mm × 3.25 mm × 0.76 mm.

2.2. Pore-size distribution

Given the 3D binary representation of pores on a rectangular grid, their morphological size distribution can be measured using tools of mathematical morphology [15], i.e. erosion and dilation. To determine the proportion of pores smaller than a given radius r a sphere of radius r is approximated on the rectangular grid and placed at each location x within the total volume X . Regarding the subset of pore voxels, $Y \subset X$, the pore space is eroded in a first step by the sphere B_x also referred to as the structuring element:

$$Y_e = \{x : B_x \subset Y\} = Y \ominus B, \quad (1)$$

meaning that the eroded set Y_e encloses all pore voxels where the sphere fits completely into the pore space. In a second step the eroded set is dilated using the same structuring element:

$$Y_d = \{x : B_x \cap Y_e \neq \emptyset\} = Y_e \oplus B. \quad (2)$$

This erosion followed by a dilation removes all pores smaller than r , the volume of which is obtained by the difference between Y and Y_d . The application of structuring elements of increasing radii r is an efficient

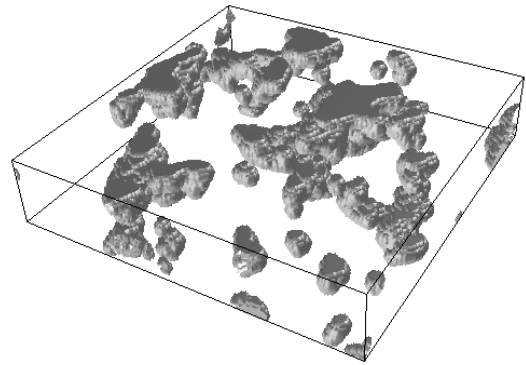


Fig. 3. 3D-reconstruction of pores >0.14 mm within a subsample of 3.25 mm × 3.25 mm × 0.76 mm.

method to determine the pore-size distribution. Fig. 3 shows the 3D-reconstruction of pores larger than 0.14 mm within the sample shown in Fig. 2. Beside the efficiency, the major advantage of this method is that the size criteria used here reflects our idea of the hydraulic diameter of a pore, i.e. the shape of the water menisci at each place within the pore space. The pore-size distribution of the undisturbed silty top soil is shown in Fig. 4. Thereby, the pore space is divided into 10 pore-size classes between 0.04 and 0.4 mm in diameter where the increments are determined by the size of the structuring elements.

2.3. Pore topology

The connectivity of the pore space has proved to play an important role in soil hydraulic properties and in its hysteretic behavior, however, a quantitative morphological description of the connectivity of the complex porous structure in soil is difficult. This is due to the facts that:

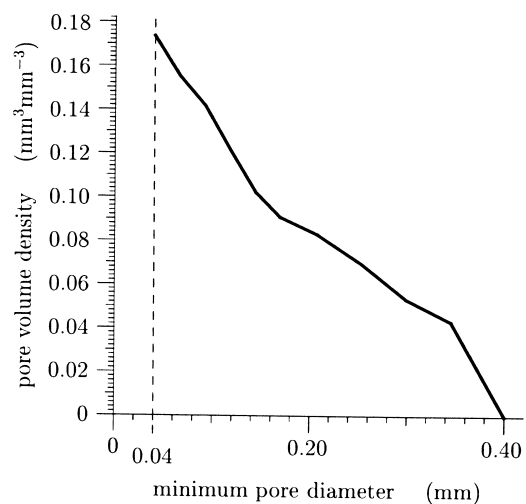


Fig. 4. Cumulative morphological pore-size distribution of undisturbed soil. Only pores >0.04 mm are considered. The total porosity is 0.503.

1. A 3D description is required which is true for any topological measure of a 3D-object.
2. The shape of the pores can be very anisotropic meaning there are pores which are very small in one or two dimensions while very large in another, i.e. cracks or root channels. Consequently, very large 3D samples have to be investigated to evaluate connectivity, but at the same time the spatial resolution must be very high to identify the pores.
3. In soil, pores of a broad spectrum of size and shape are interconnected which again requires a large quotient between sample size and resolution.
4. The sample is always limited, meaning that there are uncertainties in the evaluation of connectivity of pores at the boundaries of the sample.

To overcome most of these difficulties, the Euler number χ is a suitable measure, because an unbiased estimation is possible for a 3D cutout of arbitrary shape and volume V . The definition of the specific Euler number,

$$\chi_V = \frac{N - C + H}{V}, \quad (3)$$

is based on the fundamental topological properties, which are the number of isolated objects N , the number of redundant connections or loops C often referred to as connectivity or genus and the number of completely enclosed cavities H corresponding to hollow spheres.

In case the structure is partitioned into convex volume elements each defined by a number of vertices, edges and faces, the Euler number may be calculated according to the classical Euler formula of graph theory

$$\chi = \# \text{vertices} - \# \text{edges} + \# \text{faces} - \# \text{volumes}, \quad (4)$$

where $\#$ means ‘number of’. This implies that an estimation of χ_V is possible for any 3D binary image independent of its size and shape. The minimum 3D image is a cutout of a cube including $2 \times 2 \times 2$ voxel. In fact, an unbiased estimation of χ_V can be obtained by the frequency distribution of different voxel configurations of such $2 \times 2 \times 2$ -cubes within a 3D binary image. The theoretical background can be found by Serra [15].

However, χ does not lead to an unequivocal description of the topology, since the absolute values of N , C and H are unknown. This is the price which has to be paid for a local estimation of topological properties. Moreover, χ_V provides just a single number, describing the overall topology of the structure, it decreases with increasing connectivity. Vogel [11] introduced a connectivity function which is defined as the specific Euler number in dependency of the minimum pore size considered (Fig. 5).

This function provides quantitative information on the connectivity within and between different classes of pore size. Considering only pores larger than some 0.14 mm in diameter (Fig. 3) we realize a significant number of pores which are isolated. These pores are connected by smaller pores and may be represented by pore bodies

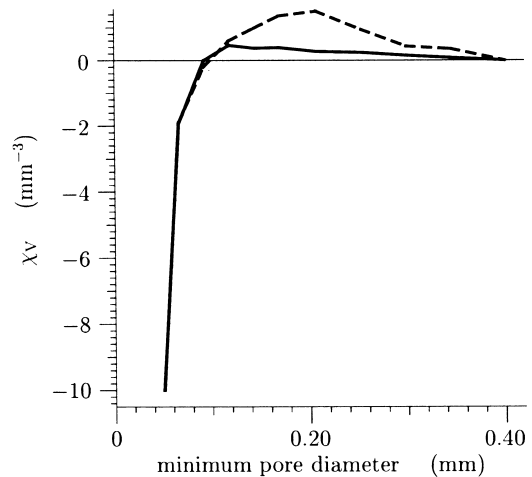


Fig. 5. Connectivity function of undistributed soil (solid line) and hypothetical connectivity function (dashed line).

in a network model. At the same time, however, there exists a connected fraction of the same pore-size class. The resulting Euler number is slightly positive (Fig. 5). By adding successively smaller pores to the structure the connectivity increases and χ_V decreases to a value of about -10 for pores larger than 0.04 mm. A striking property of the measured connectivity function is that χ_V takes only very low positive values even for the largest pore-size classes. This indicates that also large pores form continuous paths.

To evaluate the quality of the estimation of χ_V its variance was investigated in the different samples. It came out that a substantial advantage of measuring χ_V is the insignificance of the 3D shape of the sample. In the extreme just one pair of serial sections, a so-called disector [16], could be used to estimate χ_V . In fact, for technical reasons, the shape of the samples used in the present study was rather flat. The volume sampled by the serial sections was partitioned into smaller subsamples of different sizes and shapes. The different methods of partitioning are illustrated in Fig. 6. The additivity property of the Euler number guarantees that the sum of χ_V for all subsamples is constant. Clearly, the variance of χ_V decreases with increasing sample volume (Fig. 7). But also the shape of the sample has a considerable effect on the variance and herewith on the quality of the estimation. Briefly, the flatter the sample, the lower the variance. This is due to the fact that the pore structure is sampled more representatively within a flat volume compared to a more isometric sample.

3. Network model

3.1. Network geometry

The aim is now to generate a network model which corresponds to the metric and topological properties

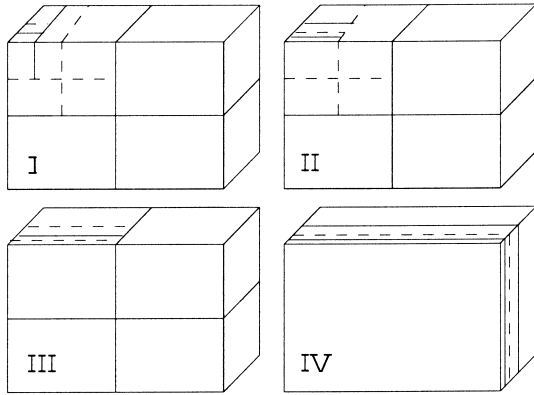


Fig. 6. Four different methods of partitioning the 3D data set to get subsamples of different shapes. The different levels of partitioning are indicated by alternating solid and dashed lines.

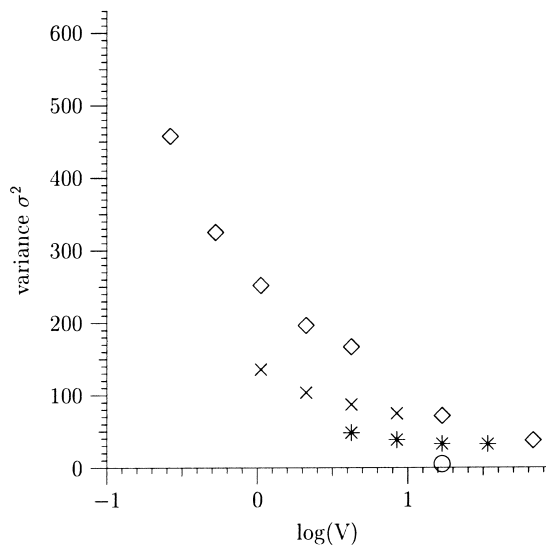


Fig. 7. Variance of estimated Euler number χ in dependency of sample volume and the shape of the sample. Method I: rhombs, method II: crosses, method III: stars, method IV: circle (see Fig. 6).

measured at the serial sections, i.e. the pore-size distribution and the connectivity function. All other properties of the network are as simple as possible so that a maximum of the complexity of the network structure is determined by directly measured morphological parameters. Therefore, a network model is chosen where the bonds are ideal cylindrical pores of a given radius and the nodes are considered to have no extra volume. The basic geometry of the network is a face-centered cubic grid with a coordination number $Z = 12$, which is the number of bonds joining at each node (Fig. 8). This number is expected to be far too high for modeling relatively large pores in soil. In previous studies [17] it was found that the coordination number of natural porous media is in the range between 2 and 5. This was confirmed by the present results on pore topology (see

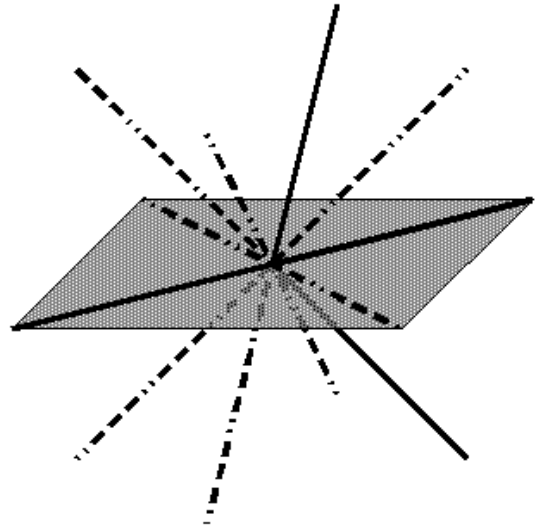


Fig. 8. Basic geometry of the face-centered cubic grid with $Z = 12$ bonds per node. Only a part ($Z_{\text{eff}} < Z$) is used to represent the measured pore space with a given pore-size distribution and topology (solid lines) the remaining pores (dashed dotted lines) represent the matrix porosity with an effective radius r_s .

below), and hence, only a small part of the 12 connections per node is used to represent pores which are between 0.04 and 0.4 mm in diameter. The remaining bonds are considered to represent smaller pores which could not be measured directly, and they are described by an effective radius $r_s < 0.02$ mm. In the simulations of flow and transport, the capillary pressure was chosen such that these small pores are always water filled. Consequently, the water phase is considered to be always continuous which is realistic in soil under such wet conditions. Moreover, the choice of an effective radius r_s representing small pores implies that the detailed geometry of these pores is of minor importance for water flow and solute transport in case the big pores are active.

Beside pore-size distribution and topology the network model has two basic parameters which are to be fixed: the grid constant λ and the effective coordination number $Z_{\text{eff}} < Z$ which determines the mean number of bonds per node used to represent the measured pore structure. These parameters can be computed from the pore-size distribution and the Euler number obtained for all pores (> 0.04 mm in this case). Given the total number of nodes N_n , the total volume of the network is

$$V = \frac{1}{\sqrt{2}} N_n \lambda^3, \quad (5)$$

and the total number of bonds is obtained by

$$N_b = \frac{1}{2} N_n Z_{\text{eff}}. \quad (6)$$

Based on the measured pore-size distribution, the probability P_i of a bond belonging to a pore-size class i is calculated as

$$P_i = \frac{1}{r_i^2} \Delta\Phi_i / \sum_j \frac{1}{r_j^2} \Delta\Phi_j, \quad (7)$$

where r_i is the radius of the pore-size class and $\Delta\Phi_i$ is the measured volume fraction of this pore-size class. The sum in (7) is for all measured pore-size classes. Then, the total volume of each pore-size class is determined by

$$V_i = N_b P_i \pi r_i^2 \lambda, \quad (8)$$

which should satisfy the condition

$$V_i = V \Delta\Phi_i. \quad (9)$$

These geometric considerations (5)–(9) lead to the relation between the grid constant λ and the effective coordination number Z_{eff} :

$$\lambda^2 = \frac{\pi}{\sqrt{2}} Z_{\text{eff}} \left[\sum_j \frac{\Delta\Phi_j}{r_j^2} \right]^{-1}. \quad (10)$$

Briefly, to satisfy a given porosity and pore-size distribution, Z_{eff} is proportional to λ^2 . To determine Z_{eff} the measurement of the Euler number χ_V including all pore-size classes can be used. According to the classical Euler formula of graph theory (4), the Euler number of the network model is obtained by

$$\chi_V = \frac{N_n - N_b}{V}, \quad (11)$$

then, using (6) and (5), we find

$$\chi_V = \sqrt{2} \frac{1 - (1/2)Z_{\text{eff}}}{\lambda^3}. \quad (12)$$

Consequently, using (10) and (12) the parameters λ and Z_{eff} can be obtained from the morphological measurements. For the given soil material, this leads to values of $\lambda = 0.247$ mm and $Z_{\text{eff}} = 2.51$.

At this point, a network may be generated which corresponds to the measured pore-size distribution. To that end, pore radii may be assigned to the bonds of the network according to the probability density defined in (7) and distributed either randomly or with a predefined autocorrelation.

The next step is to adapt the topology of the network to the measured connectivity function. The generation process of the network starts with the largest pores and the actual Euler number $\chi_{V,\text{act}}$ of the network is continuously updated using (11) while the network is generated. Prior to attributing a radius to a bond the difference between the measured Euler number of the given pore size i , $\chi_{V,i}$, and the actual Euler number of the network, $\chi_{V,\text{act}}$, is calculated. If it is negative then only an isolated bond can be chosen, otherwise only bonds connected to already existing pores are considered. In this way, the connectivity function of the network model can be adjusted to a predefined connectivity function. A wide spectrum of topological configurations can be modeled in this way, including isolated large pores

connected by narrow necks and large pores forming a continuous network. To avoid boundary effects, the network was periodic in the horizontal plane.

3.2. Simulation of hydraulic properties

To simulate the capillary pressure–saturation relation the network model is considered to be saturated with water initially. Then, the network is drained stepwise at different pressures, ψ_i corresponding to the different pore radii r_i according to the Young–Laplace equation:

$$\psi_i = 2\sigma \cos(\alpha) r_i^{-1}, \quad (13)$$

where σ is the interfacial tension between air and water, and α is the contact angle between water and solid. The latter is assumed to be zero. At each pressure ψ_i a pore is drained if (i) its radius is larger than r_i according to (13) and (ii) the pore is in contact with the non-wetting fluid. This rule is applied iteratively for each ψ_i until the network is equilibrated. Then the water content θ of the network is determined. In this way the water retention characteristic $\theta(\psi)$ is obtained.

The relative hydraulic conductivity $K_r(\psi)$ is simulated by imposing a pressure gradient Δp across the ends of the network. Water flow, q_{ij} through a cylindrical pore with radius r_{ij} connecting two nodes i and j is described by Poiseuille's law:

$$q_{ij} = \frac{\pi}{8\mu} r_{ij}^4 \frac{\Delta p_{ij}}{\lambda}, \quad (14)$$

where μ is the viscosity of water and λ is the grid constant. A further condition is given by the mass balance for each node i ,

$$\sum_j q_{ij} = 0. \quad (15)$$

This leads to a system of linear equations which is solved for the pressure distribution on all nodes in the network using the conjugate gradient method [18]. The hydraulic conductivity is then determined by the total flux through a horizontal plane. This is done for each step of capillary pressure ψ_i to get the relative hydraulic conductivity of the network.

3.3. Simulation of solute transport

Solute transport within the different network models was simulated by particle tracking for the saturated case. As for the determination of the hydraulic conductivity, a pressure gradient was imposed and the resulting flow field was calculated. Then, 10 000 particles were introduced at the nodes of the upper surface. The probability of a particle to start at a given node was proportional to the total water flux at that node. At internal nodes, the probability of a particle to move into one of the connected tubes was also proportional to the

water flux into that tube. Then, the travel time of the particles from the upper to the lower surface was recorded.

4. Results and discussion

The size of the network model was first restricted to 32^3 nodes and subsequently increased to 64^3 to investigate the effect of network size on the variability of the results. Apart from conditioning the network geometry according to the measured pore-size distribution and connectivity function, the discrete locations of the pores within the network was random. To evaluate the variability of the results due to this randomness 20 independent realizations were computed.

The resulting hydraulic properties represent a prediction based on purely morphological data. To evaluate the predictive power of the approach, the hydraulic properties were independently measured at an undisturbed soil column (diameter 16 cm, length 10 cm) in the laboratory [13]. A classical multi-step outflow experiment was performed and a parametric description of the hydraulic properties was obtained by solving the inverse problem for the Richard's equation [19].

To evaluate the sensitivity of the network behavior due to topological properties, a second set of realizations was computed assuming a hypothetical connectivity function (Fig. 5). Thereby, the Euler number of large pores was increased in comparison to the measured values. This means that large pores are more isolated and consequently less correlated.

The results for the water drainage characteristics are shown in Fig. 9. In the network model as well as in the experiment the amount of water which is lost due to decreasing capillary pressure is determined whereas the absolute water content at water saturation, i.e. the porosity, is a priori unknown. Consequently, all drainage curves were adjusted to the same value of total porosity which was calculated from bulk density of the soil column. The simulated drainage curves are quite close to the experimental results. However, large pores drain more efficiently in the network model compared to the experiment. This may be due to experimental difficulties in the saturation of the sample. Assuming entrapments of air within large pores of the experimental column would improve the agreement between simulation and experiment. The drainage characteristic of the less correlated network is clearly different. As expected the more isolated large pores within this network remain water saturated until a critical pressure is reached at which the joining smaller pores are drained.

Another interesting aspect would be to investigate imbibition in the network model. However, in contrast to drainage, the simulation of imbibition requires the calculation of the dynamics of each water–gas interface

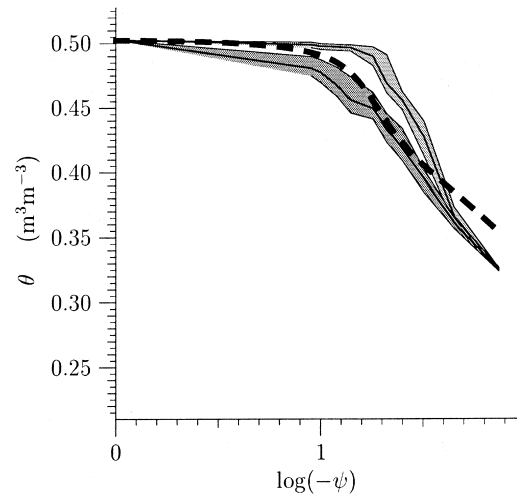


Fig. 9. Pressure saturation relation $\theta(\psi)$ of the network model with measured (dark grey) and hypothetical (light grey) topology compared to experimental results (dashed line). The grey shaded areas enclose the maximum and minimum values of 20 realizations, the size of the network was 32^3 nodes. The capillary pressure ψ is in cm.

within the network and the pressure in the non-wetting phase must be considered. These processes are highly dependent on the microscopic physico-chemical properties of the pore–solid interface which are a priori unknown.

The relative hydraulic conductivities for the different network configurations together with the experimental findings are shown in Fig. 10. Again, the simulated results are close to the experiment except in the very wet range. The difference is related to the drainage of large pores in the network model as discussed before. As for the drainage characteristic, the significance of the topological configuration is evident. A striking difference

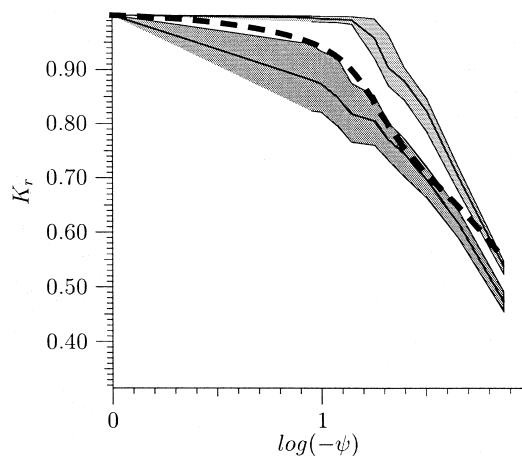


Fig. 10. Relative hydraulic conductivity $K_r(\psi)$ of the network model with measured (dark grey) and hypothetical (light grey) topology compared to experimental results (dashed line). The grey shaded areas enclose the maximum and minimum values of 20 realizations, the size of the network was 32^3 nodes. The capillary pressure ψ is in cm.

compared to the drainage curve is the large variability of the simulation results after the first steps of drainage. Obviously, the size of the network model is critical when hydraulic conductivity is considered. According to the predefined connectivity function as input to the different realizations, the percolation threshold of the pores is reached for different pore diameters or at different capillary pressures, respectively. This effect is considerably reduced by increasing the size of the network to 64^3 nodes (Fig. 11). However, increasing the size of the network does not change the average conductivities. This means that the connectivity function really quantifies the overall connectivity of the system in a sense that the number and lengths of continuous paths of a given pore size are determined by the connectivity function and do not change after further increasing of the size of the network. It is important to note that, the absolute value of hydraulic conductivity at water saturation was about one order of magnitude lower in the experiment, $K = 4.2$ cm/h, compared to the simulations, $K = 39.8$ cm/h. This is not surprising since the pores are considered to be ideal cylinders in the computational model. Considering the fact that the network simulations are only based on quantitative morphological data the prediction of the network model is good.

The results of solute transport through the network model which was adapted to the measured connectivity function is shown in Fig. 12. Thereby, the travel time of the particles is related to the expectation value $\langle t \rangle$ which corresponds to the average travel time of water through the network. Different realizations led to a high variability in small networks with 32^3 nodes (not shown). As for the hydraulic conductivity, this variability was considerably reduced by increasing the size of the network to 64^3 nodes (Fig. 12). The frequency distributions of travel times are characterized by a quick breakthrough in the simulated solute pulse and a marked tailing. A

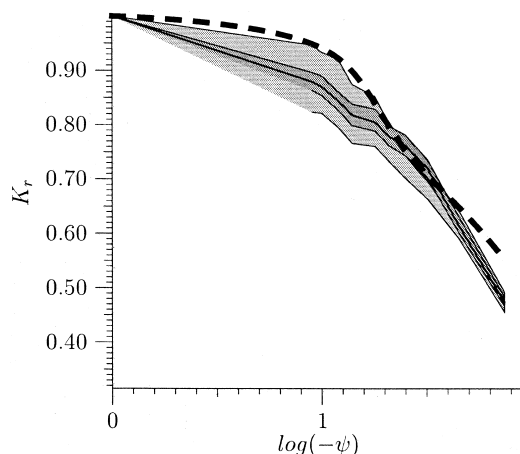


Fig. 11. Same as Fig. 10 but for a larger network with 64^3 nodes (dark grey) compared to 32^3 nodes (light grey).

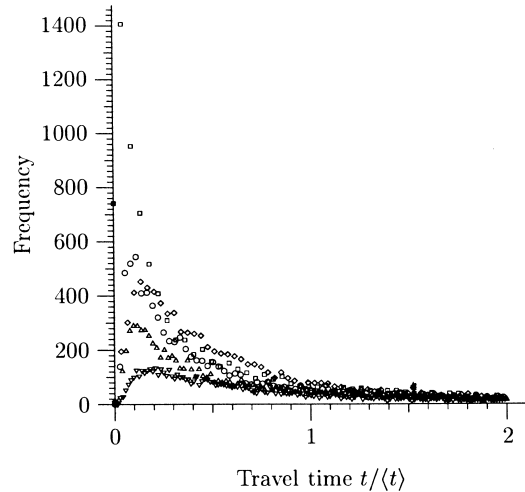


Fig. 12. Frequency distribution (five realizations) of the travel times of 10 000 particles through the network with 64^3 nodes which was adapted to the measured connectivity function. The travel time is normalized by the expectation value $\langle t \rangle$.

considerable number of particles move slowly in small pores and is contained in the long tail which is not completely covered in Figs. 12 and 13. This demonstrates that the size of the network is still far from an asymptotic limit in terms of solute transport where the frequency distribution should be Gaussian with a maximum frequency located at the expectation value. The same is true for the network with less correlated pores (Fig. 13). However, the travel time distributions are less skewed in this case. Within this less correlated structure, the increments of particle propagations are more independent which leads to a faster approach of the asymptotic state.

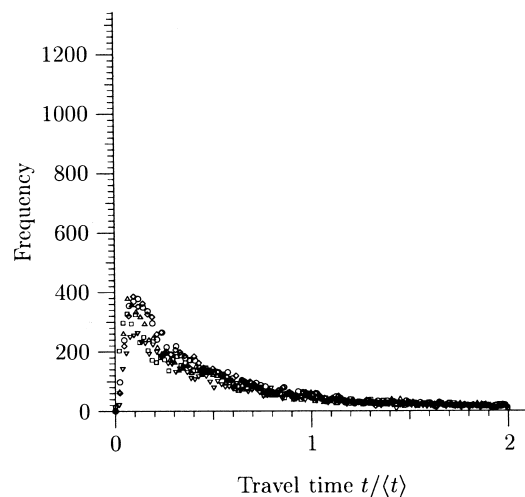


Fig. 13. Frequency distribution (five realizations with different symbols) of the travel times of 10 000 particles through the network with hypothetical topology and 64^3 nodes. The travel time is normalized by the expectation value $\langle t \rangle$.

As stated above, the connectivity function is not an unequivocal description of the topological properties of the structure. According to Eq. (3) the same Euler number can be obtained by an infinite number of combinations of the basic topological parameters N and C . Also the sizes of the different clusters of pores are not determined by the connectivity function. Given a certain combination of N and C there may be one large cluster containing all the redundant connections C and a lot of small clusters without any contribution to C . This may be critical for the behavior of the network model. To evaluate the significance of this uncertainty, an additional parameter p was introduced which controls the probability that a pore throat is attached to the largest cluster during the generation process of the network. This choice was random for the other networks. In Fig. 14 the drainage characteristics of 20 realizations with $p = 0.9$ are compared to the results of the original network. Evidently, this parameter does not affect the results which is also true for the relative hydraulic conductivities.

Another unknown parameter of the model is the effective pore radius r_s of the matrix porosity which is represented by $Z - Z_{\text{eff}}$ bonds per node. Clearly, the drainage characteristic is not influenced by this parameter within the range of capillary pressures regarded here. This is not true for the relative hydraulic conductivity, because the matrix porosity determines the conductivity of the background of the larger pores. As shown in Fig. 15 the effect of r_s on the relative conductivities increase as the relative contribution of the matrix porosity increases. Water flow near saturation is not affected by the choice of this parameter. Here, water

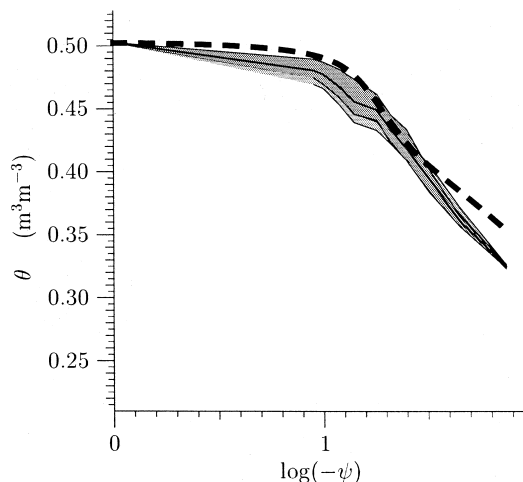


Fig. 14. Pressure saturation relation of the network model with random size distribution of pore clusters (dark grey) and weighted size distribution ($p = 0.9$, light grey) compared to experimental results (dashed line). The grey shaded areas enclose the maximum and minimum values of 20 realizations. ψ is in cm.

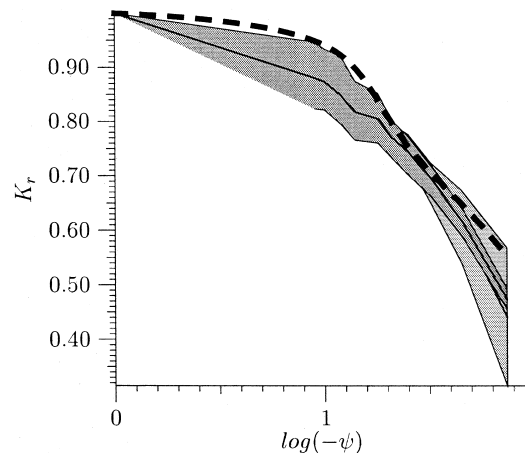


Fig. 15. Relative hydraulic conductivity of the network model with different effective radii r_s of matrix porosity: $r_s = 0.015$ mm (20 realizations, dark grey); $0.02 > r_s > 0.01$ (light grey). ψ is in cm.

flow is governed by the large pores which justify the substitution of the detailed geometry of small pores by an effective parameter.

5. Conclusions

The simulation results confirm the hypothesis that soil hydraulic properties are mainly governed by pore-size distribution and topological characteristics of the pore space. The methods used to quantify pore morphology in terms of pore-size distribution and topology proved to be suitable. In particular, the connectivity function based on the Euler number is an efficient tool which overcomes typical difficulties in measuring topological properties. Although no unequivocal description of topology is achieved the topological information provided by the connectivity function appears in this work to be sufficient. Moreover, the same concept can be easily applied to both complex pore geometry and network models.

The simple network model proposed here can be adjusted to a predefined connectivity function. Isolated macropores which are typical for network models having bodies and throats as well as continuous macropores can be represented. Additionally, small matrix pores which cannot be considered explicitly by a network model can be included by an effective parameter. This is important in modeling soil where these matrix pores guarantee the continuity of the water phase and cannot be ignored. On the other hand, the detailed geometry of this 'matrix-porosity' is of minor importance for the behavior of the porous medium. This is true for high water saturation where the hydraulic properties are governed by the large pores which are explicitly represented in the network.

The simulations of drainage, hydraulic conductivity and solute transport in networks of different sizes demonstrated that the choice of a representative volume of the structure is different for the different properties. While a relatively small network (32^3 nodes) is sufficient to describe the drainage characteristic, a much larger volume is required to achieve low variances for the hydraulic conductivity. In this work, given the topology of pores >0.04 mm of a silty soil, the maximum network size of 64^3 nodes was far too small to reach a convection–dispersion type of solute transport. However, the approach of network modeling may provide valuable information on the characteristic length scale of solute transport.

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